1. Investment Appraisal

1.1. The concept of investment appraisal

- Investment is the purchase of capital goods in expectation of future profits. All businesses make investment decisions, involving strategic decisions like relocation or adopting new technology.
- **Investment appraisal** assesses an investment decision's profitability, typically using quantitative techniques to compare future returns to costs. Managers also consider nonfinancial factors through qualitative appraisal.
- Business investments involve capital expenditure as an initial cash outflow, with expected cash inflows over the asset's useful life. Quantitative appraisal methods compare these outflows to expected inflows.

Quantitative investment appraisal: what information is necessary?

- **When appraising the profitability of investment projects using quantitative techniques, the following information is needed:**
  - Initial capital costs (e.g., buildings and equipment)
  - Estimated lifespan of asset
  - Residual value at the end of the asset's useful life (leading to further cash inflow)
  - Forecasted net cash flows (expected returns minus operating costs)
- Financial data is rarely certain, so quantitative techniques depend heavily on the reliability of estimates and forecasts

Forecasting cash flows in an uncertain environment

- **Investment appraisal techniques require forecasts of future cash flows, called forecasted net cash flows. It is assumed that:**
  - Cash inflows are equal to annual revenues earned from the project
  - Cash outflows include the initial capital cost and annual operating costs
- Forecasting cash flows is challenging and rarely 100% accurate, especially for long-term investments that require forecasts several years ahead. External factors may reduce forecast accuracy. **For example, when appraising a new airport's construction, revenue forecasts may be impacted by external factors, such as:**
  - An economic recession could reduce business and tourist traffic
  - Rising oil prices might increase air travel costs, lowering demand
  - A new high-speed rail link could shift some travelers to rail transport
- Future uncertainties are inevitable in investment appraisals, and managers must consider the risk of unforeseen events impacting cash flow forecasts. All investment decisions carry risk, so the key question is whether future profits will justify these risks.

1.2. Basic methods: Payback, Accounting Rate of Return (ARR)

- **The primary quantitative methods used in investment appraisal are:**
  - payback period
  - accounting (or average) rate of return (ARR).

The meaning, calculation, and interpretation of the payback method

- If a project costs **2,000,000 dollars** and pays back **500,000 dollars** annually, the payback period is four years, which can then be compared with alternative investments. It is common to refer to "year 0" as when the investment is made, with a negative cash flow shown in brackets (shown in table below)
- The table below shows forecasted annual **net cash flows** and **cumulative cash flows**, which represent the **running total**. The cumulative cash flow becomes positive in year 3, indicating the initial capital cost is paid back during that year. Assuming cash flows are evenly distributed, payback occurs at the end of the fourth month of year 3.
- We know this because, at the end of year 2, **50,000 dollars** is still needed to repay the initial investment. With an expected **150,000 dollars** in year 3, **50,000 dollars** represents one-third of this, equating to four months. Use this formula to find the exact month:
  - $\frac{\text{additional net cash inflow needed}}{\text{annual cash flow in year 3}}$ x 12 months
  - $\frac{50000}{150000}$ x 12 = 4 months

Year	Annual net cash flow ($)	Cumulative cash flows ($)
0	(500000)	(500000)
1	300000	(200000)
2	150000	(50000)
3	150000	100000
4	100000 (including residual value)	200000

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Why is the payback of a project important?

- **Managers can rank projects by comparing their payback periods and may set a cut-off period, rejecting any project exceeding it (e.g., five years). Reasons for setting a cut-off period include:**
  - If the investment is financed through loan, a longer payback period increases interest cost
  - Even with internal financing, capital has an opportunity cost, so quicker payback means quicker capital for other projects
  - A longer payback period increases uncertainty of the investment, as changes in the external environment over time may impact profitability.
  - Some managers that are risk-averse prefer quicker paybacks to reduce uncertainty
  - Future cash flows have lower real value due to inflation, so the faster the payback, the higher its real value is

Evaluation of the payback method

- The payback method is often used as a quick check on a project's viability or for comparing projects but is rarely used alone without other appraisal methods.

Advantages of payback method	Disadvantages of payback method
Quick and easy to calculate	Does not measure a project's overall profitability and ignores cash flows after the payback period, so a quick return might not lead to further cash inflows
Results are easy to understand	Focusing on short-term returns may lead businesses to reject highly profitable investments that take longer to repay
Emphasising quick cash returns allows focus on more accurate short-term profitability forecasts	Does not account for the timing of cash flows within the payback period, a concept further clarified by discounting in later appraisal methods
Results help identify or eliminate projects with substantially delayed returns	
Especially useful for businesses prioritising liquidity over total returns	

The meaning, calculation, and interpretation of the Accounting rate of return (ARR)

- The **accounting rate of return (ARR)**, or **average rate of return**, helps compare investments.
- For example, if Project A averages an 8% return per year and Project B 12%, choosing between them becomes easier.
- **ARR (%) is measured by this formula:**
  - ARR = $\frac{\text{average annual profit}}{\text{average investment}}$ x 100
- **where, average investment =**
  - $\frac{\text{initial capital cost} + \text{residual capital value}}{2}$

- The table below shows expected net cash flows from investing $8 million in fuel-efficient vehicles. Years 1–3 reflect annual cost savings, and year 4 includes the vehicles' residual value.

Year	Net Cash Flow (NCF)
0	($8 million)
1	$3 million
2	$3 million
3	$3 million
4	$3 million (including 2M residual capital value)

- The table below shows the five stages of calculating ARR

Stage	Working
1 Add positive net cash flows	= $12M
2 Subtract investment cost	= $12M - 8M = 4M (total profit or total net cash flow)
3 Divide by lifespan	= $4M/4 = 1M (average annual profit)
4 Calculate average investment	= $8M + 2M÷2 = 5M
5 Divide the result in stage 3 by average investment x 100 to calculate ARR%	= $1M ÷ 5M x 100 = 20%

Why is the ARR of a project important?

- **This result indicates that the business can expect an average annual return of 20% on the investment, which can be compared to:**
  - ARR on other projects
  - The minimum return the business requires, known as the criterion rate (e.g., if the business requires at least 18%, this project meets the criterion)
  - The loan interest rate, as taking a loan is not worthwhile if ARR is lower than the interest rate

Evaluation of average Rate of Return

- ARR is a popular measure for appraising projects, however is best considered together with payback results
- The two results allow consideration of both profit and cash flow timing.

Advantages of ARR	Disadvantages of ARR
It includes all cash flows, unlike payback method	It ignores cash flow timing, so two projects with similar ARR may differ in payback speed
It emphasises profitability	It includes all cash inflows, including later ones that may be less accurate
The result is clear and easy to compare with other projects	It disregards the time value of money since cash flows are not discounted (inflation)
It can be quickly evaluated against the business's criterion rate	

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1.3. Discounted cash flow method: Net Present Value (NPV)

- Managers may struggle to choose between projects if the two methods give conflicting results.
- Managers need a method that accounts for both cash flow size and timing. This can be achieved through discounted cash flow calculations.
- Ignoring inflation, a rational person would prefer **1,000 dollars** today over **1,000 dollars** in a year.
- **Today's payment is preferred for three reasons:**
  - It can be spent immediately, providing benefits instantly.
  - Saving it at the current interest rate results in more than $1,000 in a year.
  - Cash today is certain, while future payments are uncertain.
- **Discounting adjusts future cash flows to their value in today's terms, based on the interest rate**
  - If **1,000 dollars** today grows to **1,100 dollars** in a year at *10% interest*, then **1,100 dollars** in one year is worth the same as **1,000 dollars** today. **This is called the present-day value of future cash.**
- Discounting allows comparison for investment projects by considering the present value of their returns.

Discounting: How is it done?

- **The present-day value of a future sum depends on two factors:**
  - Higher interest rates reduce the value of future cash.
  - The longer the wait, the lower the value of future cash today.
- Interest rates and time are used to calculate discount factors, which can be found in discount tables (table below). To find present-day values, multiply the appropriate discount factor by the future cash flow.

Year	6%	8%	10%	12%	16%	20%
1	0.94	0.93	0.91	0.89	0.86	0.83
2	0.89	0.86	0.83	0.79	0.74	0.58
3	0.84	0.79	0.75	0.71	0.64	0.58
4	0.79	0.74	0.68	0.64	0.55	0.48
5	0.75	0.68	0.62	0.57	0.48	0.40
6	0.71	0.63	0.56	0.51	0.41	0.33

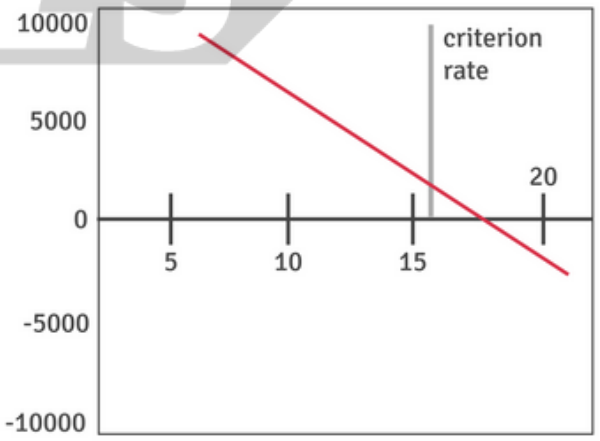
- For example, **3,000 dollars** is expected in three years at a 10% interest rate. The discount factor for Year 3 is 0.75, meaning **1 dollar** in three years is worth **75 cents** today. Multiplying **3,000 dollars** by the discount factor gives a present-day value of **2,250 dollars**.

The meaning, calculation, and interpretation of Net Present Value (NPV)

- **This investment appraisal method uses discounted cash flows. NPV is calculated by subtracting the capital cost from the total discounted cash flows. The three stages are:**
  - Multiply net cash flows by discount factors (cash flows in year 0 are not
  - discounted).
  - Add the discounted cash flows. Subtract the capital cost to find the NPV.

Year	Net cash flow	Discount factors @ 8%	Discounted cash flows
0	($10000)	1	($10000)
1	$5000	0.93	$4650
2	$4000	0.86	$3440
3	$3000	0.79	$2370
4	$2000	0.74	$1480

- The working is shown in the table above. The initial cost is paid in year 0 and is not discounted. NPV is calculated as:
  - Total discounted cash flows = $11940
  - Original investment = ($10 000)
  - Net present value = $1940
- The NPV of **1,940 dollars** means the project earns that amount in today's money values. If the finance can be borrowed at an interest rate under 8%, the investment will be profitable. Raising the discount rate reduces the NPV, as future cash flows are worth less when discounted at a higher rate. The discount rate choice is crucial for project assessment.
- Typically, businesses choose a discount rate reflecting the cost of borrowing. Even with internal finance, the interest rate should be used due to the opportunity cost of not earning interest. Alternatively, a business may use a cut-off or criterion rate to decide if the NPV is positive and the investment is viable.



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Evaluation of the NPV

Advantages of NPV	Disadvantages of NPV
Considers both the timing and size of cash flows	Complex to calculate and explain, especially for non-numerate managers
Discount rate can be adjusted for changing economic conditions, such as rising interest rates	Result depends on the discount rate used, and interest rate expectations can be inaccurate
Accounts for the time value of money and the opportunity cost of money	NPVs can be compared only if projects have the same initial capital cost, as the method does not provide a rate of return

1.4. Investment Appraisal Decisions

- Major investment projects are assessed financially and quantitatively, but other factors also play a crucial role in decision-making
- Earlier cards covered the three key quantitative appraisal methods, focusing on capital recovery speed and investment profitability. Businesses often set criteria that future projects must satisfy before approval
- These include:
  - Payback within three years
  - Minimum 15% accounting rate of return
  - Net present value at least 15% of initial capital

Qualitative factors and their impact on investment decisions

- Investment appraisal techniques provide numerical data important for decision-making
- However, managers must also consider factors that cannot be easily measured numerically but have a significant impact on a decision.
- Such qualitative factors include:
  - **Impact on the environment and the local community:** Environmental groups and bad publicity can dissuade projects in sensitive areas, prioritising CSR over profits
  - **Refusal of planning permission:** Local authorities may block or require revisions to protect community interests
  - **Aims and objectives of the business:** Investments conflicting with goals, like personal customer service, may be avoided despite potential savings
  - **Impact on the workforce:** Automation plans might be reconsidered to maintain employee relations
  - **Acceptability of risk:** Risk acceptance of managers varies, influenced by past experiences and potential failure

Comparison of investment appraisal methods

	Payback	ARR	NPV
What is measured?	Time to repay the initial capital	Annual profit as a percentage of average capital	present-day value of all net cash flows
When is the result is most useful	When there is limited finance or high interest rates	when comparing with the criterion rate, with returns from other projects and with financing cost (interest rate)	when assessing if the project returns when cash flows are discounted and when comparing with other projects of similar capital cost
Main drawbacks	Does not calculate investment profit, does not give present-day value to cash flows	Ignores timing of cash flows, does not give present-day value to cash flows	Choosing the right discount rate can be difficult, especially with fluctuating interest rates. It is hard to compare projects with vastly different initial costs

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